

Order Inventory Application

Application Endpoints

Customer Management APIs

Domain focus: Customer master data (PII-sensitive, finance/insurance aligned)

Endpoints

- **GET** /api/customers
- **GET** /api/customers/{customerId}
- **POST** /api/customers
- **PUT** /api/customers/{customerId}
- **DELETE** /api/customers/{customerId}

Specialized Queries

- **GET** /api/customers/search?email={email}
- **GET** /api/customers/{customerId}/orders
- **GET** /api/customers/{customerId}/shipments

Key Notes

- Enforce **unique email constraint**
 - Handle **PII validation**
 - Read-only links to Orders & Shipments
-

Product & Inventory APIs

Domain focus: Product catalog + stock control (core risk in enterprise apps)

Product APIs

- **GET** /api/products
- **GET** /api/products/{productId}
- **POST** /api/products
- **PUT** /api/products/{productId}
- **DELETE** /api/products/{productId}

Inventory APIs

- **GET** /api/inventory
- **GET** /api/inventory/{inventoryId}
- **POST** /api/inventory
- **PUT** /api/inventory/{inventoryId}
- **DELETE** /api/inventory/{inventoryId}

Inventory Filters

- **GET** /api/inventory/store/{storeId}
- **GET** /api/inventory/product/{productId}

Key Notes

- Enforce (store_id, product_id) unique constraint
 - Validate stock cannot be negative
 - Used downstream by Orders
-

Store Management APIs

Domain focus: Store/channel master (physical + digital)

Endpoints

- **GET** /api/stores
- **GET** /api/stores/{storeId}
- **POST** /api/stores
- **PUT** /api/stores/{storeId}
- **DELETE** /api/stores/{storeId}

Store-linked Data

- **GET** /api/stores/{storeId}/inventory
- **GET** /api/stores/{storeId}/orders
- **GET** /api/stores/{storeId}/shipments

Key Notes

- Enforce **unique store name**

Order Inventory Application

- Validate **either physical OR web address exists**
- Optional logo upload support

Order Management APIs

Domain focus: Core transactional workflow (financial-grade domain)

Order APIs

- **GET** /api/orders
- **GET** /api/orders/{orderId}
- **POST** /api/orders
- **PUT** /api/orders/{orderId}
- **DELETE** /api/orders/{orderId}

Order Queries

- **GET** /api/orders/customer/{customerId}
- **GET** /api/orders/store/{storeId}
- **GET** /api/orders/status/{status}
- **GET** /api/orders/date-range?from=&to=

Order Status Lifecycle

- **PATCH** /api/orders/{orderId}/status

Allowed status values: OPEN → PAID → SHIPPED → COMPLETE OPEN → CANCELLED PAID → REFUNDED

Key Notes

- Enforce **CHECK constraint on order_status**
- Coordinate with Inventory & Order Items
- Financial transaction integrity

Order Items & Shipment APIs

Domain focus: Fulfillment, logistics & line-level pricing

Order Items APIs

- **GET** /api/orders/{orderId}/items
- **POST** /api/orders/{orderId}/items
- **PUT** /api/orders/{orderId}/items/{lineItemId}
- **DELETE** /api/orders/{orderId}/items/{lineItemId}

Item Constraints

- Unique (order_id, product_id)
- Quantity > 0
- Unit price immutable after shipment

Shipment APIs

- **GET** /api/shipments
- **GET** /api/shipments/{shipmentId}
- **POST** /api/shipments
- **PUT** /api/shipments/{shipmentId}
- **DELETE** /api/shipments/{shipmentId}

Shipment Filters

- **GET** /api/shipments/customer/{customerId}
- **GET** /api/shipments/store/{storeId}
- **GET** /api/shipments/status/{status}

Shipment states: CREATED → SHIPPED → IN-TRANSIT → DELIVERED

Key Notes

- Shipment creation links to Orders & Order Items
- Status transitions are controlled operations